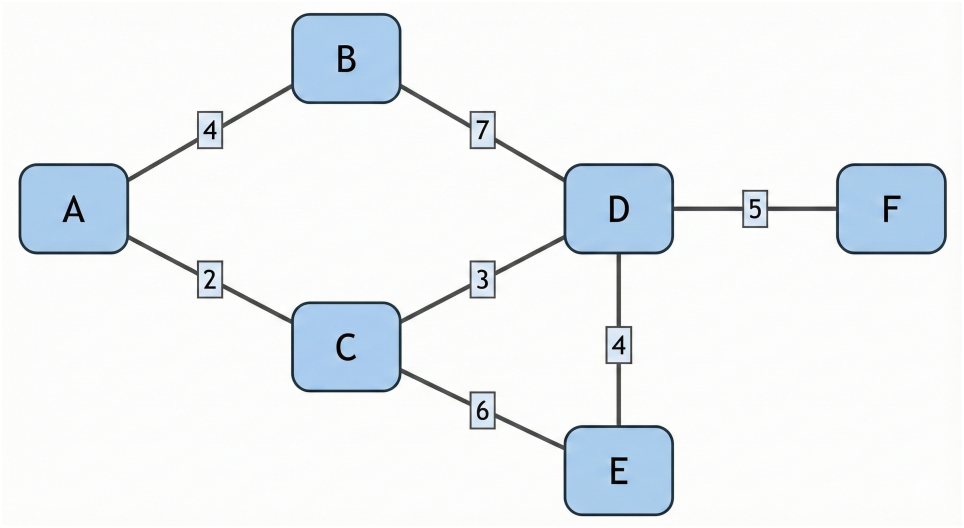
RUNI – Algorithms – Homework 6

Due date: Sunday 28/12/2025

NOTE: the general homework guidelines published in homework 1 are valid for all homework assignments in this course.

Note: Some of the questions are marked with ♠. It is recommended to work on them after the lecture and recitation of next week (we will cover some of the material only then).

Problem 1 (25 pts)



Answer the following questions (no need to justify your answers):

1. What is the weight of every MST in the graph?
2. How many different MSTs does the graph have?
3. Kruskal's algorithm is performed on the graph. What are the Union/Find sets after the three lightest edges are considered? break ties arbitrarily.

4. ♠ Assume Prim's algorithm is performed starting from vertex A. Show the best-edge array after two edges are added to the tree.

vertex	a	b	c	d	e	f
to						
cost						

5. Will your answers to (3) or (4) change if the weight of the edge CD would be 20 instead of 3?

Problem 2 (25 pts.)

Definition: Given an undirected connected graph G=(V,E), and its spanning tree, T, a vertex v is a HUB if its degree in T is identical to its degree in G.

Let G=(V,E) be a simple undirected connected graph with positive weights on the edges. Given are two vertices $a,b \in V$.

a. Suggest an algorithm for finding a minimum spanning tree $T_1=(V,E_1)$ in which a,b are HUBs (if no such MST exists, the algorithm should announce it). Justify shortly the correctness of your algorithm (no need to prove formally or to analyze time complexity).

b. Suggest an algorithm for finding a spanning tree $T_2=(V,E_2)$ whose weight is minimal among all spanning trees in which a,b are HUBs (if no such ST exists, the algorithm should announce it). Justify shortly the correctness of your algorithm (no need to prove formally or to analyze time complexity). Note that T_2 is not necessarily an MST of G.

Problem 3 (25 pts)

Let $G = (V, E)$ be an undirected connected weighted graph.
For a spanning tree T of G, let $MaxCost(T)$ be the maximum weight of an edge in T.
An *economic tree* R of a graph G is a spanning tree of G with

$$MaxCost(R) = \min(\{MaxCost(T): T \text{ is a spanning tree of } G\}.$$

Prove or give a counter example for each of the following claims:

1) If T is an MST of G, then T is an *economic tree* of G.

2) If T is an *economic tree* of G, then G is an MST of G.

♠ **Problem 4** (25 pts.)

Let G=(V,E) be a connected undirected graph with positive integer edge costs, $c: E \to N^+$.
Assume that for two vertices u,v it holds that $(u, v) \notin E$.
Suggest an efficient algorithm that given a value x, returns whether the cost of an MST of G will reduce if an edge (u,v) whose cost is x is added to the graph.
The time complexity of your algorithm should be O(|E|).
Describe the algorithm, analyze and justify the time complexity, and explain its correctness shortly (no need to prove formally).